Open problems on Diophantine m-tuples and elliptic curves

Andrej Dujella

August 22, 2026

As an addition to the book *Diophantine m-tuples and Elliptic Curves*, we give a selection of open problems and challenging questions. Some are notoriously hard problems, like the question of (un)boundedness of ranks of elliptic curves over $\mathbb{Q}$. Nevertheless, we tried to include certain variants or special cases that might be more tractable. The questions are separated with respect to the corresponding chapters, although there are certainly some questions suitable for more than one chapter.

1 Problems for Chapter 1

Problem 1.1. Find all extensions of the rational Diophantine quadruple

$$\left\{\frac{1}{16}, \frac{33}{16}, \frac{17}{4}, \frac{105}{16}\right\}$$

to a rational Diophantine quintuple.

Known extensions are $\frac{549120}{10201}$, $-\frac{26880}{177241}$. Is there any other possibility?

Problem 1.2. Is there any positive integer $k \geq 2$ such that the Diophantine quadruple $\{k-1, k+1, 4k, 16k^3-4k\}$ has more than one extension to a rational Diophantine quintuple?

Stoll (Acta Arith. **190** (2019), 239–261) proved that the extension is unique for $k = 2, 3, 4, 5$.

Problem 1.3. Let $\{a, b, c\}$ be a $D(-1)$-triple and d a positive integer such that $ad+1$, $bd+1$, and $cd+1$ are perfect squares. Do all such quadruples satisfy the formula

$$(d+a+b-c)^2 = 4(ab-1)(cd+1)\,?$$

Fujita (Acta Arith. **128** (2007), 349–375; Rocky Mountain J. Math. **39** (2009), 1907–1932) and He and Togbé (J. Number Theory **131** (2011), 120–137) obtained the positive answer for $D(-1)$-triples of the form $\{1, 2, c\}$, $\{F_{2k+1}, F_{2k+3}, F_{2k+5}\}$, and $\{1, k^2+1, k^2+2k+2\}$, respectively.

Problem 1.4. Do there exist distinct positive integers $a_1, a_2, a_3, b_1, b_2, b_3$ with the property that $a_ib_j + 1$ are perfect squares for each $1 \le i, j \le 3$?

> In the terminology of Batta, Hajdu and Pongrácz (J. London Math. Soc. (2) **111** (2025), Paper No. e70163), the problem asks whether $K_{3,3}$ is a Diophantine graph. Note that $K_{2,t}$ is Diophantine for every t, while K_5 is not Diophantine by the theorem of He-Togbé-Ziegler.

Problem 1.5. Is there any triple a_1, a_2, a_3 of distinct positive integers such that $(a_1x + 1)(a_2x + 1)(a_3x + 1)$ is a perfect square for at least six positive integers x?

> An example by Petričević (2023) shows that $(x + 1)(55x + 1)(276x + 1)$ is a perfect square for at least the following five positive integers: $x = 4, 8, 17, 89, 4870844$. Other known examples with at least five solutions are $(a_1, a_2, a_3) = (1, 2, 53)$, $(1, 5, 26)$, $(1, 10, 1407)$, $(1, 11, 316075)$, $(1, 24, 97)$, $(1, 84, 10333)$, $(5, 24, 3749)$, $(1, 571, 27409)$ (Petričević (2023,2026)), $(1, 8, 28440881)$ (Dujella (2026)), $(3, 19, 7858099)$ (Sadowski (2026)). It is not known whether there exist infinitely many such examples.
>
> There are certain infinite families with at least four positive integer solutions, including those with $(a_1, a_2) = (1, 2)$ (Sadowski (2026)), $(a_1, a_2) = (1, 8)$, $(a_1, a_2) = (k^2 - 2, k^2 - 1)$ for $k \ge 2$, $(a_1, a_2, a_3) = (a, 3a + 1, 24a^2 + 12a + 1)$, where $6a + 1 = p^2$ and $3p^2 + 5 = 2q^2$ (Dujella (2026)), $(a_1, a_2, a_3) = (1, k^2+2, k^8+4k^6+5k^4+k^2-1)$ (Sadowski (2026)), $(a_1, a_2, a_3) = (1, k^4 - 3k^2 + 1, k^6 - 7k^4 + 13k^2 - 3)$ (Andrašek (2026)).
>
> The first infinite family with five solutions found by Christopher Sadowski (2026): $a_1 = 1$, $a_2 = 12t + 7$, $a_3 = 12t^2 + 19t + 8$ with $x = t$, $x = (21t + 13 - 3s)/2$, $x = 4t + 1$, $x = (21t + 13 + 3s)/2$, $x = (16t^2 + 20t + 5)(144t^3 + 276t^2 + 157t + 24)$, where $s^2 = 33t^2 + 46t + 17$. From $(t, s) = (4, 27)$ and $(t, s) = (47, 274)$, we recover the previously known examples with five solutions $(1, 55, 276)$ and $(1, 571, 27409)$, while the next pair $(t, s) = (9932, 57059)$ gives $(a_1, a_2, a_3) = (1, 119191, 1183924204)$ with five solutions $x = 9932, 18704, 39729, 189881, 222743219182025507229956$.

Problem 1.6. Prove unconditionally that the size of a set of positive integers with the property that the product of any two of its distinct elements is 1 less than a perfect power is bounded by an absolute constant.

> Luca (Glas. Mat. Ser. III **40** (2005), 13–20) proved this result under the assumption that the *abc*-conjecture is valid.

Problem 1.7. Let $D_{m,n}(N)$ denote the number of $D(n)$-m-tuples with elements in the set $\{1, 2, \dots, N\}$ for a given positive integer N.

(a) If n is a perfect square, is there a constant $C(n)$ depending only on n, such that $D_{3,n} \sim C(n)N \log N$? Is it true that $C(n) = C(1) = \frac{3}{\pi^2}$?

(b) If n is not a perfect square, is there a constant $C(n)$ depending only on n, such that $D_{3,n} \sim C(n)N$?

(c) If n is a perfect square, what can be said about the asymptotic behaviour of $D_{4,n}$?

(d) If n is not a perfect square, is it true that $D_{4,n}$ is finite?

An affirmative answer to question (b) in the case when $|n|$ is prime or $n = -1$ is given in Adžaga, N., Dražić, G., Dujella, A., Pethő, A.: Asymptotics of $D(q)$-pairs and triples via L-functions of Dirichlet characters. Ramanujan J. **66**, Article 11 (2025).

Questions (a) and (b) are completely solved in Badesa, J.: On the asymptotics of $D(n)$-pairs and triples. Ramanujan J. **67**, Article 101 (2025). The answers to both questions are affirmative. In particular, for n a perfect square, $D_{3,n} \sim \frac{3}{\pi^2} N \log N$.

An alternative proof of (a) and (b) is given in the recent paper Dražić, G., Kazalicki, M., Mrazović, R.: Equidistribution of Diophantine pairs among the equivalence classes of quadratic forms. Preprint (2025). arXiv: 2512.22902.

Problem 1.8. Characterize elements z of the ring $\mathbb{Z}[\sqrt{-2}]$ for which there exists a $D(z)$-quadruple.

Partial results on (non)existence of $D(z)$-quadruples in $\mathbb{Z}[\sqrt{-2}]$ were obtained by Abu Muriefah and Al-Rashed (Math. Commun. **9** (2004), 1–8), Franušić (Math. Commun. **9** (2004), 141–148), Dujella and Soldo (An. Ştiinţ. Univ. "Ovidius" Constanţa Ser. Mat. **18** (2010), 81–98), and Soldo (Miskolc Math. Notes **14** (2013), 265–277), and more recently by Dujella and Soldo (arXiv:2607.07838).

Problem 1.9. Formulate a conjecture concerning connections between the existence of $D(n)$-quadruples and representability of the element n as a difference of two squares in an arbitrary commutative ring with unity, which will be in agreement with all known results and examples on that topic.

Problem 1.10. Define an analogue of a $D(n)$-m-tuple in a non-commutative ring and prove some non-trivial results concerning the existence of $D(n)$-quadruples, e.g. in the ring of 2×2 matrices with integer entries.

Dujella and Franušić (arXiv:2604.23404) proposed considering the Jordan product $A \circ B = \frac{1}{2}(AB + BA)$. The notion of regular triples naturally generalizes in this situation: if $A \circ B = R^2$, then $C = A + B + 2R$ satisfies $A \circ C = (A + R)^2$ and $B \circ C = (B + R)^2$.

Problem 1.11. Is there any $D(1)$-quintuple in the ring of integers of an imaginary quadratic field?

Problem 1.12. Is the size of $D(1)$-tuples in the ring of integers of a real quadratic field bounded by an absolute constant?

Problem 1.13. Let $\mathbb{K}$ be a number field. Is the size of $D(1)$-tuples in the ring of integers of $\mathbb{K}$ bounded by a constant depending only on the degree $[\mathbb{K} : \mathbb{Q}]$?

Problem 1.14. Find an upper bound for the size of $D(n)$-tuples in the ring $\mathbb{Z}[x]$ of polynomials with integer coefficients, which depends only on the degree of n. In particular, find an absolute upper bound in the case of cubic polynomials.

Problem 1.15. Is there any strong Diophantine quadruple, i.e. a set of four non-zero rationals $\{a_1, a_2, a_3, a_4\}$ such that $a_i a_j + 1$ are perfect squares for all $1 \leq i, j \leq 4$?

> There are examples of "almost" strong Diophantine quadruples, where all conditions for a strong quadruple are satisfied except that the condition that $a_3 a_4 + 1$ is a square is not satisfied, e.g. $\{\frac{140}{51}, \frac{2223}{3046}, \frac{278817}{33856}, \frac{3182740}{17661}\}$ (Dujella-Petričević, Experiment. Math. 17 (2008), 83–89), but it is not known whether there are infinitely many such examples.

2 Problems for Chapter 2

Problem 2.1. Are there elliptic curves over $\mathbb{Q}$ of arbitrarily large rank?

> The current record over $\mathbb{Q}$ is rank at least 29 (Elkies-Klagsbrun, 2024). Heuristics of Park, Poonen, Voight and Wood (2019) suggest that ranks over $\mathbb{Q}$ might be bounded.

Problem 2.2. Is there an elliptic curve over $\mathbb{Q}$ with torsion group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$ and rank equal to 4?

> There are 34 known curves with torsion group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$ and rank equal to 3: `https://web.math.pmf.unizg.hr/~duje/tors/z2z8.html`

Problem 2.3. Are there infinitely many elliptic curves over $\mathbb{Q}$ with torsion group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$ and rank ≥ 2? The same question for torsion groups $\mathbb{Z}/9\mathbb{Z}$, $\mathbb{Z}/10\mathbb{Z}$, and $\mathbb{Z}/12\mathbb{Z}$.

> There are infinitely many curves with rank ≥ 1 and torsion groups $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$ (Atkin & Morain (1993), Kulesz (1998), Lecacheux (2002), Campbell & Goins (2003), Rabarison (2008)), $\mathbb{Z}/9\mathbb{Z}$ (Atkin & Morain (1993), Kulesz (1998), Rabarison (2008), Gasull, Manosa & Xarles (2010), Youmbai (2024)), $\mathbb{Z}/10\mathbb{Z}$ (Atkin & Morain (1993), Kulesz (1998), Rabarison (2008), Voznyy (2021)), and $\mathbb{Z}/12\mathbb{Z}$ (Suyama (1985), Kulesz (1998), Rabarison (2008), Halbeisen, Hungerbühler, Shamsi Zargar & Voznyy (2021)).

Problem 2.4. Are there infinitely many elliptic curves over $\mathbb{Q}$ with the torsion group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$ (respectively $\mathbb{Z}/8\mathbb{Z}$) and rank ≥ 4?

> There are infinitely many curves with rank ≥ 3 and torsion groups $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$ (Dujella & Peral (2013), Dujella, Kazalicki & Peral (2021)) and $\mathbb{Z}/8\mathbb{Z}$ (Dujella & Peral (2012), Dujella, Kazalicki & Peral (2021)).

Problem 2.5. Propose a heuristic to predict the distribution of parities of ranks within known families of elliptic curves over $\mathbb{Q}$ with the torsion group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$ (respectively $\mathbb{Z}/8\mathbb{Z}$) and rank ≥ 3.

> Certain experimental data concerning the parity of ranks in some of these families can be found in Dujella, A., Kazalicki, M., Peral, J.C.: Elliptic curves with torsion groups $\mathbb{Z}/8\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$. Rev. R. Acad. Cienc. Exactas Fis. Nat. Ser. A Math. RACSAM **115**, Article 169 (2021).

Problem 2.6. Are there infinitely many elliptic curves over $\mathbb{Q}$ with torsion group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$ (respectively $\mathbb{Z}/6\mathbb{Z}$) and rank ≥ 6?

> There are infinitely many curves with rank ≥ 5 and torsion groups $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$ and $\mathbb{Z}/6\mathbb{Z}$ (Eroshkin (2009)).

Problem 2.7. Is there an infinite family of elliptic curves, parametrized by rational points of a genus 1 curve, with torsion group $\mathbb{Z}/4\mathbb{Z}$ and rank ≥ 7?

There is an elliptic curve over $\mathbb{Q}(t)$ with torsion group $\mathbb{Z}/4\mathbb{Z}$ and rank 6 (see Dujella, A., Peral, J.C.: An elliptic curve over $\mathbb{Q}(u)$ with torsion $\mathbb{Z}/4\mathbb{Z}$ and rank 6. Rad Hrvat. Akad. Znan. Umjet. Mat. Znan. **28**, 185–192 (2024).

Problem 2.8. Is there an elliptic curve over a quadratic field with torsion group $\mathbb{Z}/13\mathbb{Z}$, $\mathbb{Z}/15\mathbb{Z}$ or $\mathbb{Z}/18\mathbb{Z}$ and rank ≥ 3?

For all 26 admissible torsion groups for an elliptic curve over a quadratic field, there is a curve with that torsion and rank ≥ 2. For all of them, except possibly $\mathbb{Z}/13\mathbb{Z}$, $\mathbb{Z}/15\mathbb{Z}$ and $\mathbb{Z}/18\mathbb{Z}$, there is a curve with rank ≥ 3 (see `https://web.math.pmf.unizg.hr/~duje/tors/torsquad.html`).

Problem 2.9. Is there an elliptic curve over a quartic field with torsion group $\mathbb{Z}/22\mathbb{Z}$ and positive rank?

Solved by Maksym Voznyy on 2024/01/04.
There is a curve with torsion group $\mathbb{Z}/22\mathbb{Z}$ and rank 2 over the quartic field defined by the polynomial $f(x) = x^4 + \frac{50}{27}x^3 + \frac{281}{243}x^2 + \frac{68}{243}x + \frac{2}{81}$. The curve has the equation $y^2 - (201204z^3 + 328374z^2 + 159732z + 8019)xy - (14996874942306z^3 + 23680026731268z^2 + 10746821159640z + 931263196176)y = x^3 - (1435244994z^3 + 2266248132z^2 + 1028502360z + 89124624)x^2$, where z is a root of f.

3 Problems for Chapter 3

Problem 3.1. Is there any rational Diophantine sextuple with at least two integer elements?

> There are infinitely many rational Diophantine sextuples with at least one integer element. One such sextuple is $\{34440, 23001/280, 840/68921, 1121/11480, 19383/1640, 2530/287\}$ found by Piezas (2019).

Problem 3.2. Is there any rational Diophantine septuple?

Problem 3.3. Are there infinitely many almost rational Diophantine septuples, i.e. sets of seven rationals such that all products increased by 1 are squares, with one exception?

> Gibbs (2016) and Dujella-Kazalicki-Petričević (2018) found several examples of almost rational Diophantine septuples. For example, in $\{243/560, 1147/5040, 1100/63, 7820/567, 95/112, 38269/6480, 196/45\}$, the only missing condition is that $38269/6480 \cdot 196/45 + 1$ is not a square.

Problem 3.4. Give an explicit absolute bound for the size of rational Diophantine tuples (assuming some plausible conjectures).

> Lang's conjecture on varieties of general type implies an absolute bound for the size of rational Diophantine tuples, but it does not give an effective numerical bound in the form desired here.

Problem 3.5. Is there any elliptic curve over $\mathbb{Q}$ induced by a rational Diophantine triple with rank greater than 12?

Problem 3.6. Is there any elliptic curve over $\mathbb{Q}(t)$ induced by a family of rational Diophantine triples with rank greater than 6?

Problem 3.7. Are there infinitely many elliptic curves over $\mathbb{Q}$ with rank 0 induced by rational Diophantine triples with positive elements?

Problem 3.8. Is there any Diophantine triple $\{a, b, c\}$ such that the elliptic curve $y^2 = (ax + 1)(bx + 1)(cx + 1)$ has torsion group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$?

> Nikola Adžaga (arXiv:2608.01057) showed that for regular triples $\{a, b, c\}$ torsion group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$ cannot appear.

Problem 3.9. Is there any elliptic curve over $\mathbb{Q}(i)$ induced by a rational Diophantine triple with the torsion group $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$ and rank greater than 6?

Problem 3.10. Is there any elliptic curve over $\mathbb{Q}$ induced by a rational Diophantine quadruple with rank greater than 10?

Problem 3.11. Is there any rational Diophantine sextuple which contains a strong Diophantine triple?

> By the recent paper Dujella, A., Kazalicki, M., Petričević, V.: Rational Diophantine sextuples with strong pair. Rev. R. Acad. Cienc. Exactas Fis. Nat. Ser. A Math. RACSAM **119**, Article 36 (2025), there are infinitely many rational Diophantine sextuples which contain, e.g., a strong Diophantine pair $\{30464/2223, 22815/5168\}$.

Problem 3.12. Are there infinitely many pairs of rational Diophantine sextuples sharing a common triple such that denominators of all elements (in the lowest terms) in the sextuples are perfect squares?

> In 2024, Rathbun found two pairs of such sextuples. One such pair is
> $\{3/2^2, -12/5^2, -127500/389^2, -2093/50^2, 72512/265^2, -271859133/27970^2\}$,
> $\{3/2^2, -12/5^2, -127500/389^2, 2387/50^2, 435252/685^2, -38178413/6130^2\}$.

4 Problems for Chapter 4

Problem 4.1. Let $k \geq 2$ be an integer. Is it true that the Pellian equation

$$x^2 - (k^2+1)y^2 = k^2$$

has at most one solution with $0 < y < k-1$ and $x > 0$?

> Matthews, Robertson and White (Glas. Mat. Ser. III **48** (2013), 265–289) verified that the statement is true for $k \leq 2^{50}$, while Le and Srinivasan (Glas. Mat. Ser. III **58** (2023), 59–65) proved it for all k of the form $k = p^m q^n$, where p, q are distinct odd primes and m, n are positive integers.
>
> By recent results of Rowan Swiers (2026), obtained with the assistance of ChatGPT 5.6 Sol and Claude Fable, the statement is true for every odd k with at most three distinct prime divisors.

Problem 4.2. Is there any positive integer k, $k \neq 33539$, such that the Pellian equation

$$x^2 - (k^2-1)y^2 = 2 - k^2$$

has more than four fundamental solutions?

> For $k = 33539$ there are six fundamental solutions: $(\pm 1, 1)$, $(\pm 669941, 20)$, $(\pm 4326401, 129)$.
>
> Some information and references can be found on `http://www.numbertheory.org/keith/osipov.html`.

Problem 4.3. Are there infinitely many pairs of Diophantine quadruples with the same largest element?

> Gibbs (1999) found three such pairs:
> $\{1, 16640, 16899, 1124864520\}$ and $\{1, 399, 703920, 1124864520\}$,
> $\{120, 145673, 154155, 10778986831096\}$ and $\{8, 120, 2805566778, 10778986831096\}$,
> $\{528, 3253511, 3336933, 22929452257545400\}$ and
> $\{15, 528, 723737525421, 22929452257545400\}$. Dujella (2026) found the fourth example: $\{528, 1470, 2274769665919896, 7062345445140145061720\}$ and $\{528, 1829619951, 1827654733, 7062345445140145061720\}$.

Problem 4.4. Do there exist two Diophantine quadruples $\{a_1, b, c, d\}$ and $\{a_2, b, c, d\}$ such that $a_1 < a_2 < b < c < d$?

> It was shown by Cipu, Dujella and Fujita (Mediterr. J. Math. **19** (2022), Article 187) that there are only finitely many such examples, in fact, the largest element satisfies $d < 10^{10^{26}}$.

Problem 4.5. Is there any Diophantine quadruple $\{a, b, c, d\}$ such that $a \equiv b \equiv c \equiv d \pmod{k}$ with $k \geq 3$?

> It was shown by Dujella and Saradha (Indag. Math. (N.S.) **25** (2014), 131–136) that a positive answer to Problem 4.6 would imply a negative answer to Problem 4.5.

Problem 4.6. Are all Diophantine quadruples regular?

Problem 4.7. Are all $D(4)$-quadruples regular?

A positive answer to Problem 4.7 would imply a positive answer to Problem 4.6, while a negative answer to Problem 4.6 would imply a negative answer to Problem 4.7. Namely, $\{a, b, c, d\}$ is a regular Diophantine quadruple if and only if $\{2a, 2b, 2c, 2d\}$ is a regular $D(4)$-quadruple.

Problem 4.8. Let $\{a, b, c\}$ be a Diophantine triple. Do all integer points on the elliptic curve $y^2 = (ax + 1)(bx + 1)(cx + 1)$ (such that $x \neq -1$) satisfy the condition that $ax + 1$, $bx + 1$ and $cx + 1$ are all perfect squares?

Partial positive answers are known for the family of elliptic curves from Problem 4.9, and for families corresponding to triples of the form $\{1, 3, c\}$ (Dujella and Pethő, Publ. Math. Debrecen **56** (2000), 321–335) and $\{F_{2k}, F_{2k+2}, F_{2k+4}\}$ (Dujella, J. Théor. Nombres Bordeaux **13** (2001), 111–124).

Ana and Matej Jurasić (with a help of ChatGPT 5.6 Sol Pro) (2026) found an infinite family of counterexamples: $a = 5$, $b = 115$, $c = c_k$, $x = x_k$, where $c_k = (S_k^2 - 1)/5$, $x_k = (c_k - 120)/575$ and $T_k^2 - 23S_k^2 = -22$, $k \equiv 15 \pmod{1150}$. The smallest counterexample obtained with this constructions is $c = 19726599787079129775103071759543738099950374011720$,
$x = 34307130064485443087135776973119544521652824368$.

Problem 4.9. Let $k \geq 3$ be an integer. Is it true that all integer points on the elliptic curve $y^2 = ((k - 1)x + 1)((k + 1)x + 1)(4kx + 1)$ are given by $(x, y) \in \{(0, \pm 1), (16k^3 - 4k, \pm(128k^6 - 112k^4 + 20k^2 - 1))\}$?

The positive answer is known if the rank of the curve is equal to 1 (by Dujella (Acta Arith. **94** (2000), 87–101)) or $k \leq 5000$ (by Najman (Rocky Mountain J. Math. **40** (2010), 1979–2002)). Also if $k = 3n^2 + 2n - 2$ or $k = \frac{1}{2}(3m^2 + 5m)$ and the rank is equal to 2 (by Dujella (Acta Arith. **94** (2000), 87–101)).

Problem 4.10. Propose a heuristic to predict the rank distribution within the family of elliptic curves

$$y^2 = (F_{2k}x + 1)(F_{2k+2}x + 1)(F_{2k+4}x + 1).$$

In the range $1 \leq k \leq 100$, there are 20 values of k with rank 1, 45 values with rank 2, 21 values with rank 3, 5 values with rank between 1 and 3, 6 values with rank between 2 and 4, 2 values with rank between 3 and 5, and 1 value with rank between 2 and 6.

It seems that a reasonable heuristic is that 25% of the curves have rank 1, 50% have rank 2, 25% have rank 3, and the curves of rank ≥ 4 have density 0. Besides the usual minimalist and root-number heuristics, the main new ingredient is that for odd k there is an additional rational point $Q_k = \left(-\frac{F_{k-1}}{L_k F_{k+1} F_{k+2}}, \frac{F_{2k+1}}{L_k F_{k+1} F_{k+2}}\right)$, which is generically independent of the usual point $P_k = (0, 1)$ (except that for $k = 1$ the two points coincide). The point Q_k was found by Dujella in 2026, with assistance from ChatGPT 5.6 Sol Plus, based on computations in PARI/GP.

5 Problems for Chapter 5

Problem 5.1. Is there any $D(n)$-quadruple for $n = -3, 3, 5, 8, 12$ or 20?

Problem 5.2. Is the set of all integers n, $n \not\equiv 2 \pmod 4$, for which there are at most two distinct $D(n)$-quadruples finite?

Problem 5.3.

(a) Is there a polynomial Diophantine quadruple with the property $D((4k-1)(4k+1))$ apart from the quadruples obtained from two known polynomial Diophantine quadruples with the property $D(4k+3)$?

(b) Let $n = (4k-1)(4k+1)$, where $k > 3$ is a positive integer satisfying the conditions that $4k-1$, $4k+1$, $(n-1)/2$, $(n-9)/2$, $(9n-1)/2$, $16k-5$, $16k+5$ are all primes (examples of such numbers n are 9294986317823, 27363653936783, 48073810526783). Is there any such n for which there exist more than two $D(n)$-quadruples?

If some of the numbers $4k-1$, $4k+1$, $(n-1)/2$, $(n-9)/2$, $(9n-1)/2$ is not prime, then additional $D(n)$-quadruples (with positive elements) can be obtained from Dujella, A.: A problem of Diophantus and Dickson's conjecture. In: Győry, K., Pethő, A., Sós, V.T. (eds.) Number Theory, Diophantine, Computational and Algebraic Aspects, pp. 147–156. Walter de Gruyter, Berlin (1998), while if $16k-5$ or $16k+5$ is not prime, then there are additional $D(n)$-quadruples with mixed signs (Dujella (2026)).

Problem 5.4. Let n be a non-zero integer. Prove that the size of a $D(n)$-tuple is bounded above by a function of the form $c \log\log |n|$ for an absolute constant c.

Problem 5.5. Is there a third power (respectively a fourth power) Diophantine quadruple?

Problem 5.6. Find an absolute upper bound for the size of $D(p^2)$-tuples, where p is a prime.

Problem 5.7. Is there any rational $D(q)$-quintuple for $q = 1579$?

This value $q = 1579$ is the smallest positive integer q for which no rational $D(q)$-quintuple is known.

Problem 5.8. Does a rational $D(q)$-quintuple exist for every rational number q?

Problem 5.9. Are there infinitely many rational $D(q)$-quintuples for every rational number q?

Problem 5.10. Is there any rational $D(q)$-sextuple with q which is not a perfect square?

Problem 5.11. Is there any non-zero integer n and a $D(n)$-sextuple with all odd elements?

> Petričević (2018) found examples of $D(n)$-quintuples with all odd elements: $\{-273375, -361375, -504063, 833, 1377\}$ is a $D(831406275)$-quintuple and $\{-9, 59, 6075, 47291, 555579\}$ is a $D(5117175)$-quintuple.

Problem 5.12. Are there infinitely many Diophantine triples that are also $D(n_i)$-triples for $1 < n_2 < n_3 < n_4$?

> The following triples with the given property were found by Adžaga, Dujella, Kreso and Tadić (2018): $\{4, 12, 420\}$, $\{10, 44, 21252\}$, $\{4, 420, 14280\}$, $\{40, 60, 19404\}$, $\{78, 308, 7304220\}$, $\{4, 485112, 16479540\}$, $\{15, 528, 32760\}$.
>
> New examples found by Dujella and ChatGPT (2026): $\{2070, 11928, 23936\}$ with $n_2 = 14196736$, $n_3 = 25378816$, $n_4 = 29758820281$, and $\{20, 1980, 637602\}$ with $n_2 = 327244681$, $n_3 = 1328052649$, $n_4 = 100997436001$.

Problem 5.13. For a positive integer k, denote by $n_1(k)$ the smallest (in the absolute value) non-zero integer for which there exists a triple $\{a, b, c\}$ of non-zero integers and a set N of integers, such that $|N| = k$, $n_1(k) \in N$, and $\{a, b, c\}$ is a $D(n)$-triple for all $n \in N$.

(a) Is it true that $n_1(5) = 36$?

(b) What can be said (assuming some plausible conjectures) about the asymptotic behaviour of $|n_1(k)|$ for large k?

> Sadowski (2026) answered question (a) showing that $n_1(5) \leq 4$. In fact, he showed that also $n_1(6) \leq 4$ with the following example: $\{a, b, c\} = \{30, 1056, 65520\}$, $N = \{4, 266436, 15248601, 21479364, 63936324, 1037903409\}$. Furthermore, $n_1(7) \leq 36$ with $\{a, b, c\} = \{8, 1620, 257040\}$, $N = \{36, 857529, 1470564, 200322756, 238596849, 452017161, 16308809476\}$.

Problem 5.14. Is there a set of four non-zero integers which is a $D(n_1)$-, a $D(n_2)$-, and a $D(n_3)$-quadruple for three distinct non-zero integers n_1, n_2, and n_3?

Problem 5.15. Is there a set of five non-zero integers which is a $D(n_1)$-quintuple and a $D(n_2)$-quintuple for two distinct non-zero integers n_1 and n_2?

Problem 5.16. Is there a rational Diophantine quintuple whose elements are all squares?

Problem 5.17. Are there infinitely many triples of distinct non-zero rationals (a, b, c) such that $a + 1$, $b + 1$, $c + 1$, $ab + 1$, $ac + 1$, $bc + 1$, and $abc + 1$ are all squares?

Some triples with the given property are $(8, \frac{15}{49}, \frac{1320}{49})$, $(8, \frac{312}{529}, \frac{495}{529})$, $(\frac{25}{144}, -\frac{143}{144}, \frac{1792}{3249})$. By the recent paper Dujella, A., Szalay, L.: Four squares from three numbers. Preprint (2025). arXiv: 2506.14013, there are infinitely many triples (a, b, c) of integers greater than 1 such that $ab + 1$, $ac + 1$, $bc + 1$, and $abc + 1$ are all squares.

Problem solved in the recent paper Dujella, A., Kazalicki, M., Petričević, V.: Seven squares from three numbers. Preprint (2026). arXiv:2604.08729. The solution is based on the fact that the quartic $3u^4 - 9u^2 + 7 = z^2$ has infinitely many rational points. This gives an infinite sequence of triples with given property, starting with
$(8, \frac{312}{529}, \frac{495}{529})$,
$(\frac{312}{529}, -\frac{152880}{165649}, -\frac{78374557}{87628321})$,
$(-\frac{152880}{165649}, \frac{724255280}{736742449}, -\frac{63009087694401}{122040649934401})$,
$(\frac{724255280}{736742449}, \frac{24490915482072}{12448992625969}, -\frac{4510665894525110607837}{9171701314839342058081})$,
$(\frac{24490915482072}{12448992625969}, -\frac{2539564321528123032}{5054545907282329441}, \frac{14261842404349331345950974819695}{62924004727379507987985949853329})$.

In the same paper, it is proved that there does not exist a triple of non-zero integers with the same property.

Problem 5.18. Let a, b, c be integers such that $1 < a < b < c$ and $c+1$, $ab+1$, $ac+1$, $bc+1$, $abc+1$ are all squares. Is it true that necessarily $(a, b, c) = (5, 7, 24)$?